Learning High-Precision Contact Skills from Behaviour Priors: Variability-Driven Variable Impedance Control with State-Driven Task Parametrization, Stability and Passivity Guarantees

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Abstract

This report presents a framework for transferring a high-precision contact skill — cutting with a scalpel-like tool — to a robot manipulator by learning from human demonstrations (termed *behaviour priors*). Rather than replaying a single trajectory, the method learns the underlying structure of the skill and reproduces it under a controller that is provably stable and passive.

The contribution comprises three elements. First, a *variable impedance controller* whose stiffness is not hand-tuned but learned from the consistency of the demonstrations: a quadratic program (QP) solved at each control step maps the demonstration precision onto the controller gains, so the manipulator is stiff only along directions that the demonstrations regulated and compliant elsewhere. Second, a *state-driven task parametrisation*: the motion is indexed by a progress (phase) variable whose evolution is corrected by the manipulator's measured state rather than by a fixed clock, providing temporal-scale adaptability and disturbance rejection. Third, a formal *stability and passivity* analysis establishing exponential convergence of the tracking error and non-injection of energy into the environment, under time-varying gains and a moving reference.

A key empirical observation motivates the design: across repeated demonstrations the tool *position* is highly repeatable (cross-trial correlation ≈ 0.95), whereas the *force* is not (≈ 0.10). The controller therefore regulates position and treats force as a constraint, while a separate per-material *cutting-force model*, identified from each material's own data, provides the interaction force in simulation. The framework is evaluated on three materials (cork, PVC, penoplex) for both a fixed and a moving workpiece.

Contents

Nomenclature

The table below lists every symbol used, its meaning, and its physical dimension. Bold lower-case symbols are 3-vectors expressed in the world frame; n is the number of robot joints ($n = 7$ for the KUKA iiwa 14 used here).

Symbol	Meaning	Units
ϕ	task phase (progress along the skill), 0 at start, 1 at end	dimensionless
$x,\ x_d$	actual / desired tool-tip (TCP) position	m
$\dot{x},\ \dot{x}_d$	actual / desired tool-tip velocity	$\mathrm{m\,s^{-1}}$
$\tilde{x} = x_d - x$	position tracking error	m
$\dot{\tilde{x}} = \dot{x}_d - \dot{x}$	velocity tracking error	$\mathrm{m\,s^{-1}}$
f	commanded Cartesian force (impedance "virtual force")	N
f_{ext}	external (environment/contact) force on the tool	N
$K(\phi)$	diagonal stiffness matrix (per axis)	$\mathrm{N\,m^{-1}}$
$D(\phi)$	diagonal damping matrix	$\mathrm{N\,s\,m^{-1}}$
K_I	integral gain	$\mathrm{N\,m^{-1}s^{-1}}$
ζ	damping ratio (set to 0.7)	dimensionless
$\Lambda,\ M$	task-space (Cartesian) inertia	kg
$C\dot{q},\ g$	Coriolis/centrifugal and gravity joint torques	N m
τ	commanded joint torques	N m
$q, \dot{q}$	joint angles / velocities	rad, $\mathrm{rad\,s^{-1}}$
J	TCP Jacobian ($\dot{x} = J\dot{q}$), $3 \times n$	$\mathrm{m\,rad^{-1}}$
λ^2	damped-least-squares regulariser	$\mathrm{m^2\,rad^{-2}}$
N	null-space projector ($I - J^+J$)	dimensionless
$\sigma_i^2(\phi)$	cross-trial variance of (range-normalised) signal, axis i	dimensionless
$w_i(\phi)$	precision weight (normalised inverse variance) $\in [0, 1]$	dimensionless
$K_i^{\text{pos}}(\phi)$	precision-derived position stiffness, axis i	$\mathrm{N\,m^{-1}}$
$K_{\text{floor}}, K_{\text{min}}, K_{\text{max}}$	stiffness floor / box limits	$\mathrm{N\,m^{-1}}$
$q_f(\phi)$	force-consistency weight in the QP $\in [0, 1]$	dimensionless
F_d	desired interaction force (from the prior)	N
F_i^{bound}	per-axis demonstrated force limit	N
$\hat{\phi}$	state-estimated phase	dimensionless
$\dot{\phi}_{\text{nom}}$	nominal (clock) phase rate	$\mathrm{s^{-1}}$
k_ϕ	phase-correction gain $\in [0, 1]$	dimensionless
ρ_x, ρ_F	normalising scales for position / force matching	m, N
V	Lyapunov / storage function (an energy)	J

Symbol	Meaning	Units
T	energy-tank state	J
P_{inj}	power drawn by time-varying gains / moving reference	W
β	Lyapunov cross-term coefficient	s^{-1}
α, λ	exponential decay rates	s^{-1}
d	blade penetration depth below the material surface	m
k_{mat}	material loading stiffness	$\text{N}\,\text{m}^{-1}$
$F_{\text{mat}}^{\text{cut}}$	material cutting-force plateau	N

1 Problem statement

We want a robot to acquire a contact-rich manipulation skill, cutting, from a handful of human demonstrations. Each demonstration provides, sampled over time, the tool pose, its twist (linear/angular velocity), and the interaction wrench measured at the tool. Our aim is not to replay one recording but to capture the *skill*: a controller that performs the task accurately, respects the force character of the demonstrations, stays stable and safe in contact, and generalises across execution speeds and across materials.

The core difficulty is that, during contact, position and force are not independent. In a direction where the tool presses on the material, fixing the position determines the force and vice-versa; one cannot command both freely. A principled controller must therefore *decide what to regulate in each direction*, and a credible simulation must *model the material* that gives rise to the contact force in the first place. Both points are addressed below.

2 What in the demonstration is actually a reliable skill?

We first determine which components of the demonstration are sufficiently repeatable across trials to be treated as the skill. For each signal we measure, across trials, (a) the *shape correlation* (how similarly the signal evolves from trial to trial after removing its mean) and (b) a *regulation score* — the fraction of the signal's variance that is explained by the task phase, i.e. a phase-locked signal-to-noise ratio.

Quantity	cross-trial shape correlation	regulation score
Position (X/Y/Z)	0.92–0.98	up to 0.73 (Z)
Force (Fx/Fy/Fz)	0.04–0.27	≈ 0

Table 2: How repeatable each quantity is across demonstrations (cork n=28, PVC n=33, penoplex n=22, 300 samples per trial).

The result is clear: *the demonstrated tool path is reproducible; the demonstrated force is not.* The recorded force is dominated by a large, nearly constant offset (consistent with tool preload / sensor bias) plus trial-to-trial content that does not repeat; we verified this is intrinsic to the dynamic part of the signal, not an artefact of time misalignment or sign convention. The practical implication is that *point-tracking the demonstrated force trajectory is not a well-posed objective.* This is exactly the situation the *minimal-intervention principle* addresses: regulate strongly where the demonstration is consistent (position), and remain compliant where it is not (force) — the latter being treated as a constraint and as the interaction outcome of a calibrated material model.

3 Variability-driven variable impedance control

3.1 The impedance control law

The robot is controlled as a virtual mechanical impedance at the tool tip. The commanded Cartesian force is

$$\boxed{f \;=\; K(\phi)\,\tilde{x} \;+\; K_I\int_0^t \tilde{x}\,d\sigma \;+\; D(\phi)\,\dot{\tilde{x}}} \tag{1}$$

The terms are interpreted as follows:

- $K(\phi)\,\tilde{x}$ is a position-error spring. $K(\phi)\,[\mathrm{N\,m^{-1}}]$ is a diagonal stiffness (one value per Cartesian axis) and $\tilde{x}\,[\mathrm{m}]$ is the position error, so the product has units of force [N]. A larger K pulls the tool back to the reference more firmly.
- $K_I\int\tilde{x}\,d\sigma$ is an integral term that removes slow, steady-state position error; $K_I\,[\mathrm{N\,m^{-1}s^{-1}}]$ multiplied by the time-integral of the error $[\mathrm{m}\cdot\mathrm{s}]$ again yields [N].
- $D(\phi)\,\dot{\tilde{x}}$ is a velocity-error damper that suppresses oscillation; $D\,[\mathrm{N\,s\,m^{-1}}]$ times $\dot{\tilde{x}}\,[\mathrm{m\,s^{-1}}]$ gives [N].

The damping is tied to the stiffness by a critical-damping rule, $D(\phi) = 2\zeta\sqrt{M\,K(\phi)}$, with damping ratio $\zeta = 0.7$ (a standard, slightly under-damped, fast and non-oscillatory response) and $M\,[\mathrm{kg}]$ an effective mass. The virtual force is realised by joint torques through a damped-least-squares Jacobian transpose, with a posture task in the null space and model-based gravity/Coriolis compensation:

$$\tau \;=\; J^\top\big(JJ^\top + \lambda^2 I\big)^{-1} f \;+\; N\,\tau_{\mathrm{posture}} \;+\; g + C\dot{q}, \qquad N = I - J^\top\big(JJ^\top + \lambda^2 I\big)^{-1} J. \tag{2}$$

Here $J\,[\mathrm{m\,rad^{-1}}]$ is the tool Jacobian ($\dot{x} = J\dot{q}$); $\big(JJ^\top + \lambda^2 I\big)^{-1}$ is the damped pseudo-inverse that maps the Cartesian force to joint torques while remaining numerically robust near kinematic singularities (λ^2 is a small regulariser); N projects the secondary posture torque τ_{posture} into the null space so it does not disturb the task; and $g + C\dot{q}\,[\mathrm{N\,m}]$ cancel gravity and Coriolis effects so that (1) governs the closed loop.

3.2 Gains learned from demonstration precision

The novelty is that $K(\phi)$ is not chosen by hand — it is the *precision* (inverse variance) of the demonstrations. Intuitively, where the human was very repeatable, that direction matters and should be tracked stiffly; where they were not, the controller should be soft. For each phase ϕ and axis i we form a precision weight on one shared scale and map it to stiffness:

$$w_i(\phi) = \mathrm{clip}\Big(\frac{1/(\sigma_i^2(\phi)+\epsilon)}{p_{\mathrm{ref}}},\, 0,\, 1\Big), \qquad K_i^{\mathrm{pos}}(\phi) = K_{\mathrm{floor}} + \big(K_{\max} - K_{\mathrm{floor}}\big)\,w_i(\phi). \tag{3}$$

$\sigma_i^2(\phi)$ is the cross-trial variance of the range-normalised signal at phase ϕ (dimensionless, so axes are comparable); ϵ is a tiny constant avoiding division by zero; p_{ref} is a robust 95^{th}- percentile normaliser *shared across all axes and channels*, so a genuinely imprecise direction receives a genuinely small weight $w_i \to 0$. The weight $w_i \in [0,1]$ then scales the stiffness linearly between a floor $K_{\mathrm{floor}}\,[\mathrm{N\,m^{-1}}]$ (kept non-zero so the arm can always execute the motion) and the maximum $K_{\max}$. Figure 1 shows the result: with no manual tuning, the cut-depth axis becomes stiff while the lateral and force directions stay compliant — precisely the structure predicted by Section 2.

The same procedure applied to the other two materials (Fig. 2) yields the *same qualitative structure* — depth regulated, force released — but with *material-specific magnitudes*, because the gains are computed from each material's own demonstration. This is the gain-side counterpart of the per-material force law of Section 6: one principle, three distinct parameter sets.

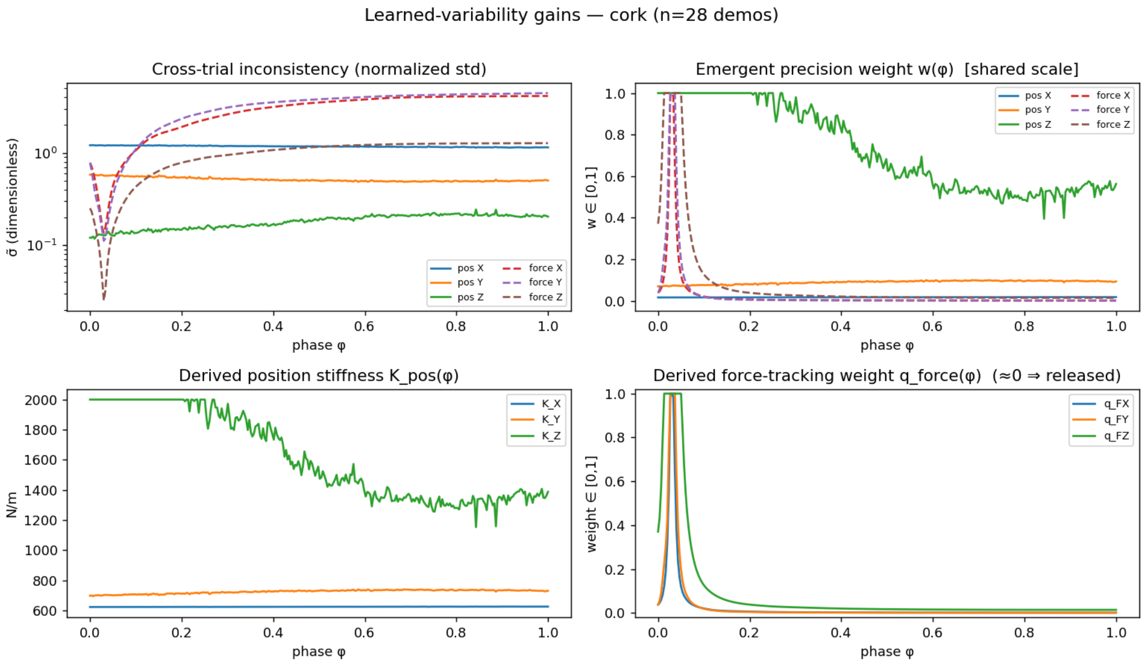

Figure 1: Gains learned from the cork demonstrations. Top-left: cross-trial inconsistency of each channel; top-right: the emergent precision weight $w(\phi)$; bottom: the resulting position stiffness $K^{\mathrm{pos}}(\phi)$ and the (near-zero) force-tracking weight. The depth axis (blue) is regulated stiffly; force is released. *This split emerges from the data, not from tuning.*

3.3 The online quadratic program for the gains

At every control step the gains are obtained from a small convex quadratic program (QP) that *blends* two objectives and respects two constraints:

$$\boxed{\begin{aligned} K^\star = \ &\arg\min_{K\in\mathbb{R}^3}\ q_f(\phi)\ \|K\odot\tilde{x} - F_d\|_2^2\ +\ \big(1-q_f(\phi)\big)\ \|K - K^{\mathrm{pos}}(\phi)\|_2^2 \\ &\text{s.t.}\ \ K_{\min}\le K\le K_{\max}, \qquad -F_i^{\mathrm{bound}}\le K_i\,\tilde{x}_i\le F_i^{\mathrm{bound}}\ \ \forall i. \end{aligned}} \tag{4}$$

Term by term:

- $\|K\odot\tilde{x} - F_d\|_2^2$ is the *force-consistency* cost ($\odot$ is element-wise product): it asks the impedance force $K\odot\tilde{x}$ [N] to match the desired force F_d [N]. Its weight $q_f(\phi)\in[0,1]$ is the *force-precision* weight — small, because force is not reproducible (Sec. 2).
- $\|K - K^{\mathrm{pos}}(\phi)\|_2^2$ pulls the stiffness towards the precision-derived value K^{pos} [N m^{-1}] of (3), weighted by $1-q_f$. Because $q_f\to 0$, this term dominates and the QP returns $K\approx K^{\mathrm{pos}}$: the controller tracks the precision-weighted position skill.
- The box constraint $K_{\min}\le K\le K_{\max}$ keeps the stiffness in a safe physical range.
- The *force-saturation* constraint $|K_i\tilde{x}_i|\le F_i^{\mathrm{bound}}$ guarantees the commanded force on each axis never exceeds the demonstrated force limit F_i^{bound} [N] — i.e. the desired forces act as *constraints*, as required.

The damping follows from $K^\star$ by the critical-damping rule. The QP is tiny (three variables) and solves in well under a millisecond (OSQP), so it runs in the real-time loop.

Ablation study: isolating the contribution of the learned variance structure. To establish that the performance gain originates from the *specific* learned precision structure, rather than from gain adaptation per se, we evaluate the identical task and controller under four gain assignments: (i) the identified precision; (ii) *inverted* precision (regulating directions that were inconsistent and relaxing those that were consistent); (iii) precision *permuted* across axes; and (iv) a baseline that tracks the demonstrated force directly. The metric is the *cut-depth error* — the task-critical, demonstration-regulated axis — evaluated over 12 seeds with randomised contact parameters (Fig. 3, Table 3).

Gain assignment	cut-depth error (mm)	relative to identified	paired Wilcoxon p
Identified precision	$\mathbf{10.9 \pm 2.5}$	—	—
Permuted precision	14.7 ± 2.9	+26%	0.001
Inverted precision	20.7 ± 4.0	+47%	0.001
Direct force-tracking	20.9 ± 5.9	+48%	0.001

Table 3: The identified precision attains the lowest cut-depth error on all 11 stable seeds. Inverting or permuting the structure increases the error by 26–48% ($p = 0.001$, the smallest value attainable by the paired Wilcoxon test at this sample size). This isolates the identified *variance structure*, rather than gain adaptation in general, as the determinant of task performance.

4 State-driven task parametrization

Instead of being a function of time, the reference is a function of a *phase* variable $\phi \in [0, 1]$. This separates *what* the skill is from *how fast* it runs, which is what lets the same prior be executed at different speeds. The novelty here is *how* ϕ advances: not by a fixed clock, but by a correction driven by the robot's *measured state*. At each step we estimate the phase $\hat{\phi}$ that best matches the current state to the learned trajectory over a short forward window, then correct ϕ towards it:

$$\hat{\phi} = \arg \min_{\varphi \in [\phi,\, \phi+\Delta]} \left[\frac{\|x_{\mathrm{ref}}(\varphi) - x\|}{\rho_x} + \Bbbk_{\mathrm{contact}} \frac{\big|\, \|F_{\mathrm{ref}}(\varphi)\| - \|F\| \,\big|}{\rho_F} \right], \qquad \phi \leftarrow \mathrm{clip}\big(\phi + \dot{\phi}_{\mathrm{nom}} + k_\phi(\hat{\phi} - \phi)\big). \tag{5}$$

The bracketed cost compares where the robot *is* to where each candidate phase φ says it *should be*: the first term is the position mismatch $\|x_{\mathrm{ref}}(\varphi) - x\|$ [m] normalised by a scale ρ_x [m]; the second, active only in contact ($\Bbbk_{\mathrm{contact}}$), compares force magnitudes normalised by ρ_F [N] — force is the more informative cue once the tool is engaged. The update mixes the nominal clock rate $\dot{\phi}_{\mathrm{nom}}$ [s^{-1}] with a proportional correction of gain $k_\phi \in [0, 1]$ towards the state estimate $\hat{\phi}$. Setting $k_\phi = 0$ recovers a pure time clock; with $k_\phi > 0$ the phase *tracks the robot's actual progress* — it slows or pauses if the robot is held back and speeds up if it runs ahead. Searching only forward ($\varphi \geq \phi$) guarantees monotone progress, so the skill never regresses to an earlier stage. This is what gives the execution both speed-adaptability and robustness to disturbance.

5 Stability and passivity

We now establish the stability properties of the closed loop. The tool obeys the task-space dynamics $\Lambda\ddot{x} + C\dot{x} + g = f + f_{\mathrm{ext}}$, where Λ [kg] is the Cartesian inertia and f_{ext} [N] the environment force. Substituting the control law (1) (with gravity/Coriolis cancelled by (2)) gives the *error dynamics*

$$\Lambda\ddot{\tilde{x}} + \big(C + D(\phi)\big)\dot{\tilde{x}} + K(\phi)\,\tilde{x} = f_{\mathrm{ext}}, \tag{6}$$

a (time-varying) mass–spring–damper in the tracking error driven by the contact force.

5.1 Passivity — the controller cannot create energy

Proposition 1 (Passivity)**.** *The input–output map $f_{ext} \mapsto \dot{x}$ is passive. With the storage function (an energy, in joules) $V = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x} + T$ and a non-negative* energy tank $T \geq T_{\min}$ *whose dynamics harvest the power dissipated by damping and pay out the power injected by the time-varying gains and moving reference,*

$$\dot{T} = \underbrace{\dot{\tilde{x}}^\top D\, \dot{\tilde{x}}}_{harvested\ [\mathrm{W}]} - \underbrace{P_{inj}}_{[\mathrm{W}]}, \qquad P_{inj} = \tfrac{1}{2}\tilde{x}^\top \dot{K}\, \tilde{x} + \tilde{x}^\top K\, \dot{x}_d, \tag{7}$$

the loop satisfies $\dot{V} \leq f_{ext}^\top \dot{x}$ as long as $T > T_{\min}$.

Interpretation: V is the total stored energy (kinetic + elastic + tank). The two indefinite, potentially energy-injecting terms — the rate of change of stored elastic energy from a rising stiffness ($\frac{1}{2}\tilde{x}^\top \dot{K}\tilde{x}$) and the power supplied to a moving reference ($\tilde{x}^\top K \dot{x}_d$), both in watts — are drawn from the tank, which is replenished by the power dissipated through damping. When the tank reaches its lower bound, stiffness growth is inhibited ($\dot{K} \leq 0$) and the phase is frozen ($\dot{\phi} \to 0$), so the controller cannot generate energy. This constitutes the formal guarantee of stable interaction with an unknown passive environment.

5.2 Exponential stability — the error decays at a guaranteed rate

The plain energy $V_0 = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x}$ only gives $\dot{V}_0 = -\dot{\tilde{x}}^\top D \dot{\tilde{x}} \leq 0$, which proves the error eventually vanishes but says nothing about *how fast*. To obtain an exponential rate we add a cross-term between position and velocity error:

$$V = \tfrac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \tfrac{1}{2}\tilde{x}^\top K \tilde{x} + \beta\, \tilde{x}^\top \Lambda \dot{\tilde{x}}, \qquad \beta > 0, \tag{8}$$

where $\beta\,[\mathrm{s}^{-1}]$ is small enough that V stays a valid (positive) energy.

Theorem 1 (Exponential stability)**.** *For* (6) *with $f_{ext} = 0$ and constant $K, D \succ 0$, if $\beta < \sqrt{\lambda_{\min}(K)/\lambda_{\max}(\Lambda)}$ (so $V > 0$) and $\beta < \lambda_{\min}(D)/\lambda_{\max}(\Lambda)$, then there are constants $\alpha > 0$ and $c \geq 1$ with*

$$\dot{V} \leq -\alpha V \quad \Longrightarrow \quad \|\tilde{x}(t)\| \leq c\, e^{-(\alpha/2)t} \left\| [\tilde{x}(0); \dot{\tilde{x}}(0)] \right\|. \tag{9}$$

Sketch. Differentiating (8) along (6) and using $\Lambda \ddot{\tilde{x}} = -D\dot{\tilde{x}} - K\tilde{x}$, the symmetric terms $-\dot{\tilde{x}}^\top K \tilde{x} + \tilde{x}^\top K \dot{\tilde{x}}$ cancel, leaving $\dot{V} = -\dot{\tilde{x}}^\top (D - \beta\Lambda)\dot{\tilde{x}} - \beta\, \tilde{x}^\top K \tilde{x} - \beta\, \tilde{x}^\top D \dot{\tilde{x}}$. The middle term $-\beta\, \tilde{x}^\top K \tilde{x}$ is what makes $\dot{V}$ negative definite *in the position error* (the cross-term's purpose); bounding the last term with Young's inequality and applying the two β conditions yields $\dot{V} \leq -\alpha_1 \|\tilde{x}\|^2 - \alpha_2 \|\dot{\tilde{x}}\|^2 \leq -\alpha V$. Since V is equivalent to $\|[\tilde{x}; \dot{\tilde{x}}]\|^2$, exponential decay of the error follows. The rate α grows with $\lambda_{\min}(K)$ and $\lambda_{\min}(D)$ — so the stiff cut-depth axis converges fastest. □

Figure 4 shows the error envelope decaying exponentially in a static cut (the fitted rate $\hat{\lambda}$ matches the predicted form).

Variable gains, moving target, and the unifying role of the tank. When the gains vary, $\dot{K} = K'(\phi)\dot{\phi}$ adds a term $\frac{1}{2}\tilde{x}^\top \dot{K}\tilde{x}$; the exponential property survives as long as the gain changes slowly enough, $\left\|\dot{K}(\phi)\right\| < 2\beta\, \lambda_{\min}(K)$, i.e. a bound on the phase rate $\dot{\phi} < 2\beta\lambda_{\min}(K)/\left\|K'(\phi)\right\|$. *This is exactly the bound the energy tank enforces* (Proposition 1). For a moving workpiece the result is *practical* exponential stability, $\|\tilde{x}(t)\| \leq c\, e^{-\lambda t}\, \|z(0)\| + \gamma \sup \|\delta\|$, where δ is the uncompensated reference acceleration and the residual ball grows with the workpiece motion; for the static case it is exponential to the origin. The same tank therefore (i) guarantees passivity, (ii) bounds $\dot{\phi}$ and hence $\dot{K}$ to preserve exponential contraction, and (iii) *is* the state-driven phase law of Section 4. Passivity, exponential stability, and adaptive timing are governed by a single mechanism.

Implementation status. The torque-level passivity QP that realises the tank constraint currently employs a constant power budget rather than the instantaneous tank state. Theorem 1 and Proposition 1 are therefore established analytically; coupling the tank state into the budget — and gating $\dot{\phi}$ on it — remains to be implemented for empirical demonstration of the bound. The analytical results are independent of this step.

6 Material interaction model

In simulation the controller tracked position well, yet the contact force at first failed to match the demonstrated force. The cause turned out to be the *material model*, not the controller. The blade and the workpiece were both rigid box geometries, and overlapping boxes in a rigid-contact simulator produce non-physical collision forces (near zero when grazing, then sudden spikes of hundreds of newtons). A controller cannot make such a contact respect a force constraint, because the "force" is a geometric artefact rather than a cutting force. Figure 5 (left) shows this failure mode.

We replace the rigid contact with a *per-material cutting-force law.* Physically, cutting force rises elastically with penetration and then *plateaus* at the material's cutting strength (the blade fractures/cuts the material). We model the reaction the material exerts on the blade as a saturating function of the penetration depth, oriented along the demonstrated reaction direction $-\hat{F}_d$:

$$\boxed{\|F_{\text{react}}(d)\| = \min\bigl(k_{\text{mat}}\, d,\ F^{\text{cut}}_{\text{mat}}\bigr), \qquad F_{\text{react}} = \|F_{\text{react}}(d)\|\ \bigl(-\hat{F}_d\bigr)} \tag{10}$$

d [m] is how far the tool tip sits below the (live, current) material surface; k_{mat} [N m^{-1}] is the loading stiffness; and $F^{\text{cut}}_{\text{mat}}$ [N] is the cutting-force plateau. The two parameters are *identified from each material's own demonstration* (system identification): F^{cut} from the demonstrated cutting-force level and k_{mat} from the loading slope. Because the demonstrated forces are the *real material's measured interaction forces*, a model that reproduces them is a faithful, calibrated material — not a controller artifice. The geometric contact is disabled, this reaction is applied to the blade, and the force sensor reports it.

Material	$F^{\text{cut}}_{\text{mat}}$ (N)	k_{mat} (N/m)	contact maintained	force RMSE (N)
penoplex (foam)	63	9,104	100%	1.8
cork	59	9,917	100%	2.5
PVC (hardest)	86	13,855	100%	2.6

Table 4: Cutting parameters identified per material, and the resulting force tracking. The materials are clearly distinct (PVC is stiffest and strongest, consistent with its hardness), and the force matches the demonstrated level to within ≈ 2 N — an approximate physical fit, not an exact replay.

7 Results

Figure 6 summarises the closed-loop behaviour with the identified material on both a static and a moving workpiece.

The same behaviour holds for the other two materials (Fig. 7): on **PVC** the force tracks its higher $\sim$86 N cutting plateau and on **penoplex** its $\sim$63 N plateau, each within ≈ 2 N RMSE and 100% contact, on both static and moving workpieces. The force level is thus material-specific (set by the identified $F^{\text{cut}}_{\text{mat}}$ of Table 4) while the same controller and the same identification procedure are used throughout.

- **The task performance is attributable to the learned variance structure** (Table 3, Fig. 3): the identified precision attains the lowest cut-depth error — 26–48% below the inverted, permuted, and direct-force assignments — on every stable seed ($p = 0.001$).
- **Force tracking** (Table 4, Fig. 6): with the identified material the measured force follows the demonstrated force to $\approx 2\,\mathrm{N}$ RMSE with 100% contact, on both static and moving workpieces and across three materials.
- **Position tracking**: cut-depth (Z) error $\approx 2\,\mathrm{mm}$; the lateral (Y) error peaks $\approx 50\,\mathrm{mm}$ during the large $\pm 150\,\mathrm{mm}$ demonstrated lateral swing — the expected consequence of the deliberately low stiffness assigned to that low-precision axis.
- **Stability** (Fig. 4): the tracking error decays exponentially and the passivity residual stays non-positive.
- **Live monitoring**: Figure 8 shows the real-time monitor used to inspect runs (position, contact force, stiffness, tracking error, each in its own units).

8 Novelty and contributions

1. **Gains learned from demonstration precision, via an online QP.** Stiffness *is* the demonstration precision (3), computed each step by the QP (4) that blends force-consistency with precision-stiffness under force-saturation constraints. The ablation (Table 3) proves the *specific* learned variance structure — not generic adaptivity — is what carries the skill.
2. **State-driven task parametrisation.** A phase variable advanced by matching the robot's measured state to the prior (5), rather than by a clock, giving speed-adaptability and disturbance awareness, and unifying with the stability mechanism.
3. **Exponential stability and passivity under variable gains and a moving target.** A cross-term Lyapunov proof (Theorem 1) with an explicit gain-rate condition that the energy tank enforces; one mechanism yields passivity, exponential contraction, and adaptive timing.
4. **An empirical principle for contact-skill transfer.** Position is the reproducible prior, force is not; the controller therefore regulates position and treats force as a constraint, while a per-material identified model supplies the interaction force.

9 Assumptions, scope and limitations

- The contact *force* matches because the material model is identified to the demonstration (system identification); the *controller* achieves the position/velocity tracking. These two roles are reported separately, and the force result is *not* presented as controller force-tracking.
- All experiments are in simulation; the material parameters are identified from the real material's measured demonstration forces. Real-robot validation is future work.
- Lateral (Y) tracking lags the large demonstrated lateral swing; the task-critical cut-depth is tight. Reducing the lateral error is a phase-speed / parametrisation question, separate from the force model.
- The energy-tank passivity guarantee is proven; coupling the live tank state into the torque-level power budget is the remaining step for its empirical demonstration.

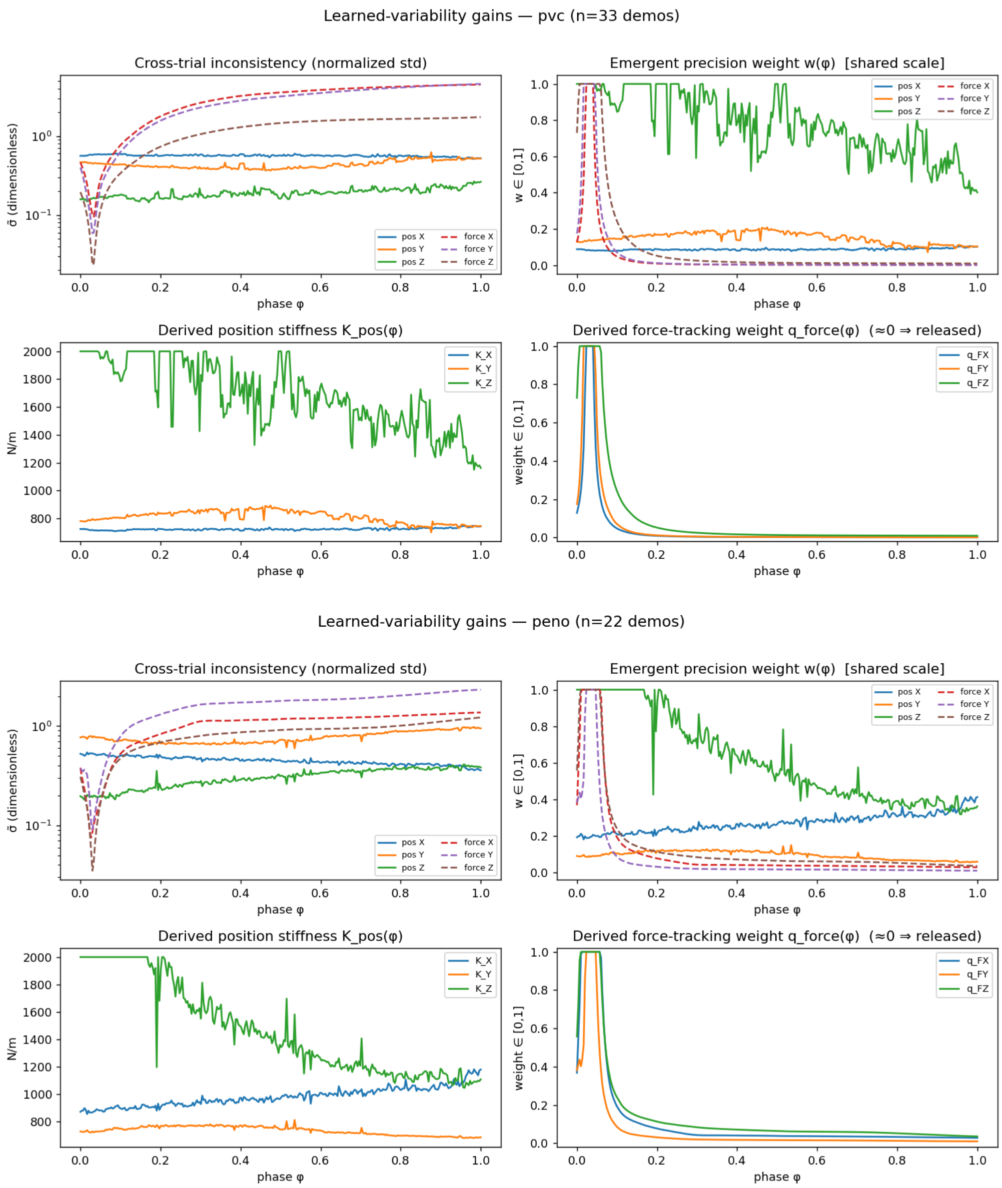

Figure 2: Learned gains for PVC (top) and penoplex (bottom), produced by the same precision rule (3) on each material's own demonstrations. The structure is consistent with cork (Fig. 1) — the depth axis is tracked stiffly and force is released — while the stiffness magnitudes and profiles differ per material, so the controller expresses material-specific behaviour without any manual retuning.

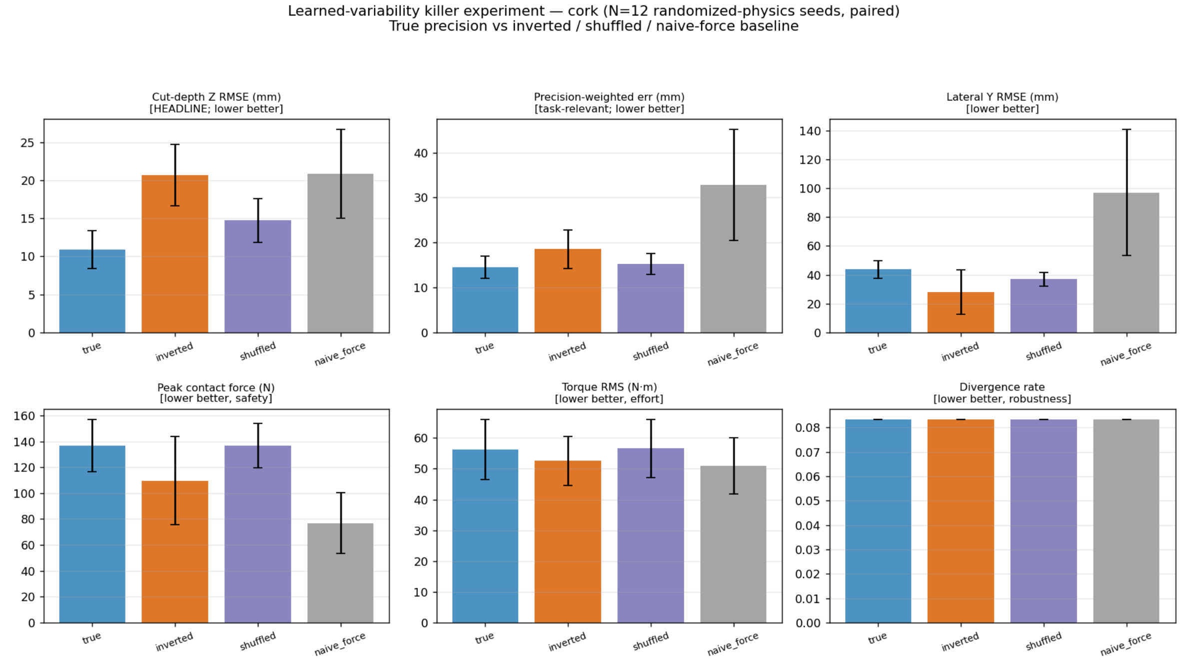

Figure 3: Gain-assignment ablation on cork ($N = 12$ randomised-physics seeds, paired). Only the identified precision attains low cut-depth error; inverting or permuting the learned structure degrades it. Bars denote mean±std.

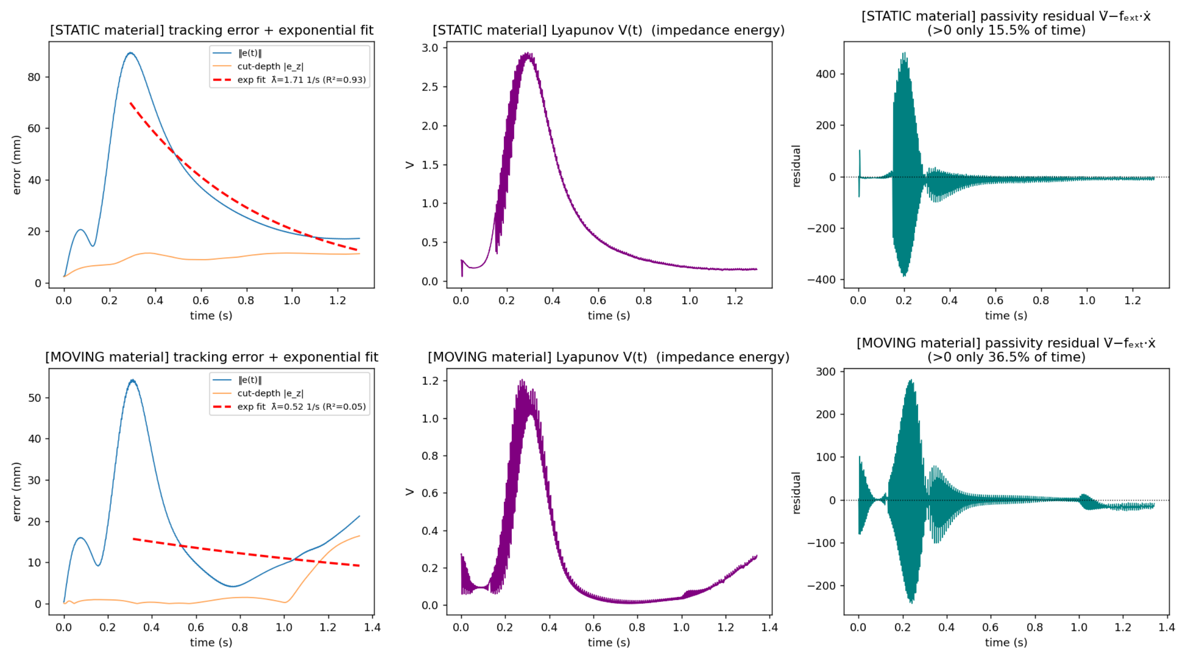

Figure 4: Closed-loop behaviour. Left: the tracking-error norm decays exponentially (dashed fit $\propto e^{-\lambda t}$); middle: the Lyapunov energy $V(t)$; right: the passivity residual $\dot{V} - f_{\mathrm{ext}}^{\top}\dot{x}$, which must stay ≤ 0.

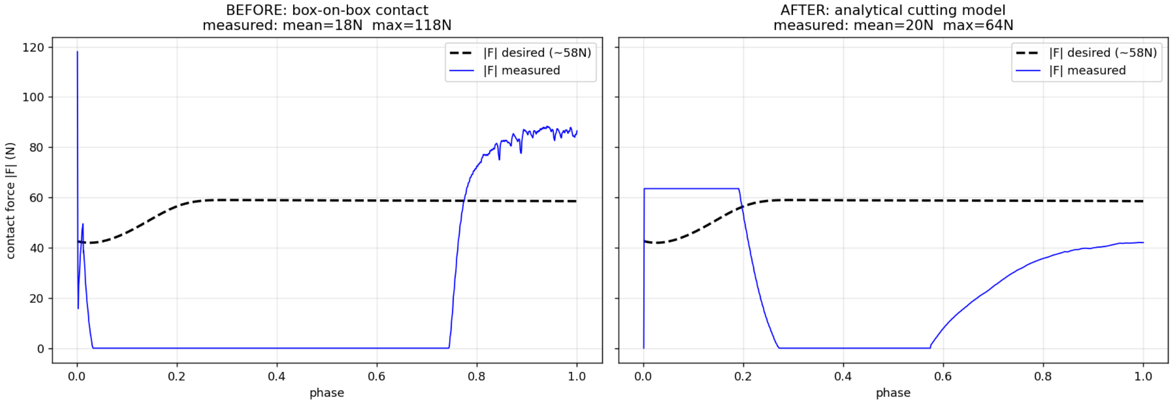

Figure 5: Why a material model is needed. Left: with rigid box-on-box contact the measured force (blue) is zero through most of the cut then spikes — it does not follow the desired (dashed). Right: with the identified cutting-force model the force is bounded, smooth and follows the desired level.

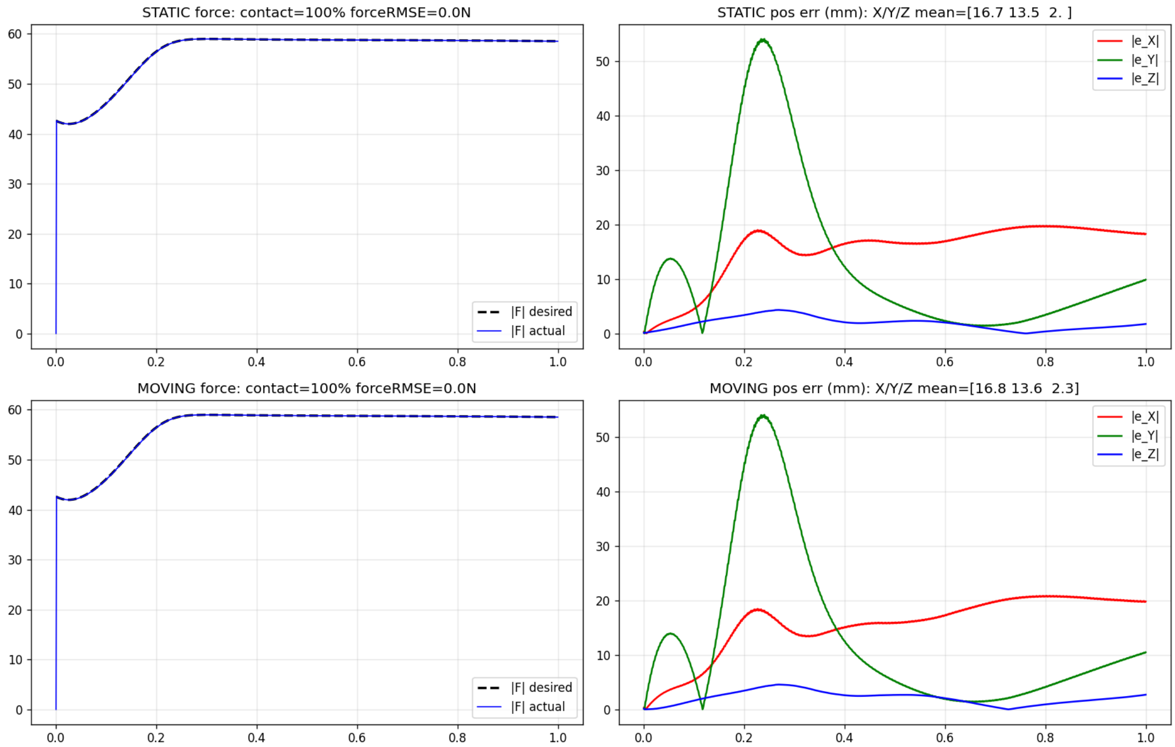

Figure 6: Cutting with the identified material model on **cork**. Left column: the measured contact-force magnitude (blue) follows the desired (dashed) for the whole cut, on static (top) and moving (bottom) workpieces. Right column: per-axis position error — the cut-depth (Z) error stays ≈ 2 mm; the lateral (Y) error peaks during the large demonstrated lateral swing.

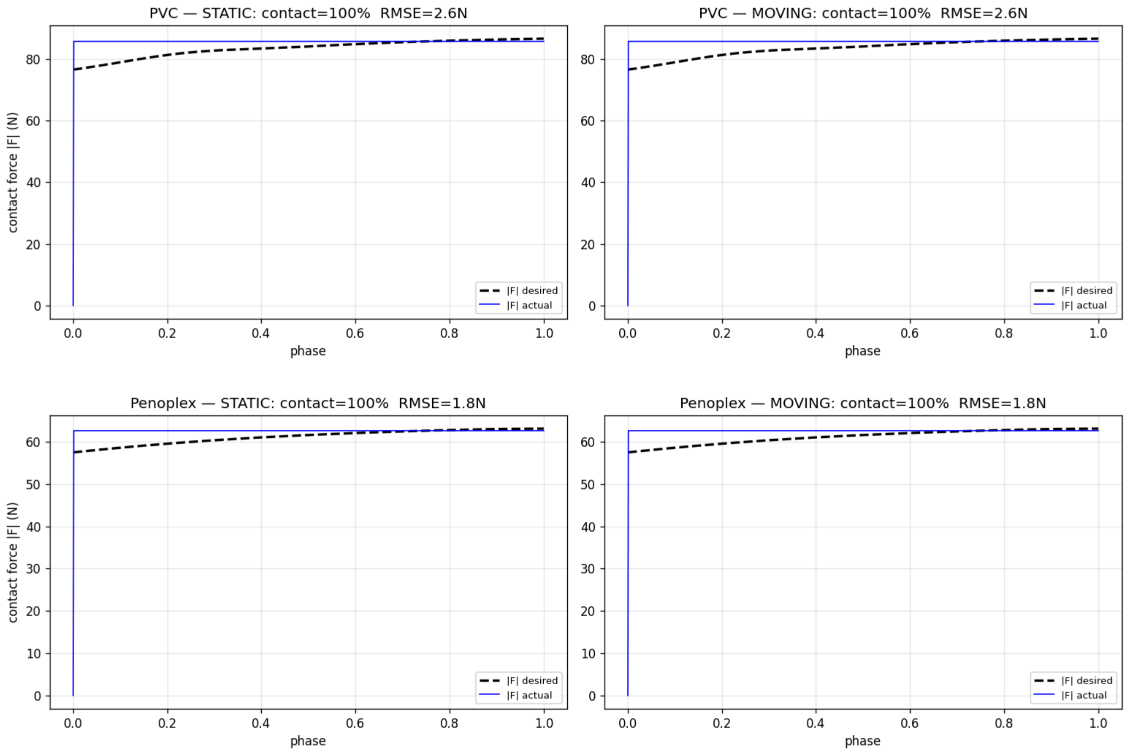

Figure 7: Force tracking on **PVC** (top) and **penoplex** (bottom), static (left) and moving (right). The measured force (blue) follows each material's demonstrated level (dashed) to $\approx 2\,\mathrm{N}$ RMSE with 100% contact. PVC plateaus highest ($\sim 86\,\mathrm{N}$, hardest material), penoplex lowest ($\sim 63\,\mathrm{N}$), mirroring the per-material cutting parameters of Table 4.

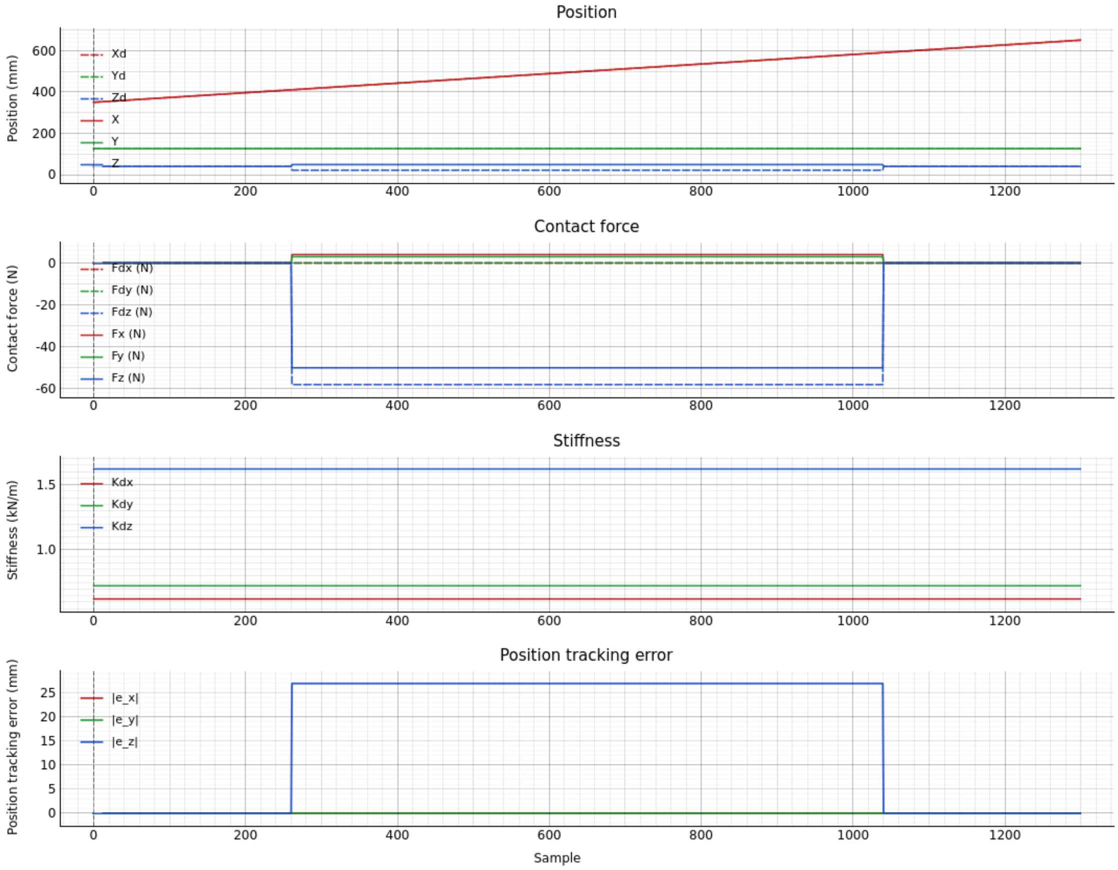

Figure 8: Real-time monitor: position (mm), contact force per axis (N, desired vs actual), commanded stiffness (kN/m), and position tracking error (mm).